

# Open-vocabulary 3D object detection with promptable segmentation

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## ABSTRACT

Three-dimensional object detection for autonomous driving is dominated by detectors trained on large corpora of human-annotated 3D boxes. Such a detector learns a fixed category list, and everything outside it is invisible. This paper asks whether the task can be solved training-free and open-vocabulary. A promptable segmentation model (SAM3), queried with class names as text prompts, supplies instance masks in the vehicle's six surround-view cameras, and the masks are turned into metric 3D boxes using the geometry of the scene. The core is a controlled three-stage comparison on nuScenes in which 2D detection is held fixed and only the source of 3D geometry changes. Geometry predicted from images alone reaches 0.183 mean average precision (mAP) under the official protocol; fitting boxes from raw LiDAR points inside the same masks with training-free rules reaches 0.298 mAP / 0.348 nuScenes detection score (NDS) at zero labeling cost; borrowing supervised box geometry at inference time lifts the same detections to 0.413 mAP / 0.555 NDS, which locates the pipeline's largest deficit in measurement precision rather than 2D detection, while class confusion and confidence calibration survive that substitution. Reversing the direction, a three-state camera-witness rule built from the same masks improves a supervised LiDAR-only detector from 0.596 to 0.630 mAP, roughly half the gain of fully supervised camera fusion, with no training. A coverage analysis shows that SAM3 finds 84% of in-range objects with a correctly named mask; the classes that fail in the official metric are misnamed or geometrically unforgiving, not unseen.

## 1. Introduction

An autonomous vehicle acts on what it perceives. Every driving decision ultimately rests on a list of the objects around the vehicle, each carrying a position, a size, an orientation and a velocity, all measured in meters relative to the vehicle. Producing this list from on-board sensors is the task of three-dimensional (3D) object detection, and an error at this stage propagates directly into behavior: an undetected pedestrian is a pedestrian the planner will not brake for.

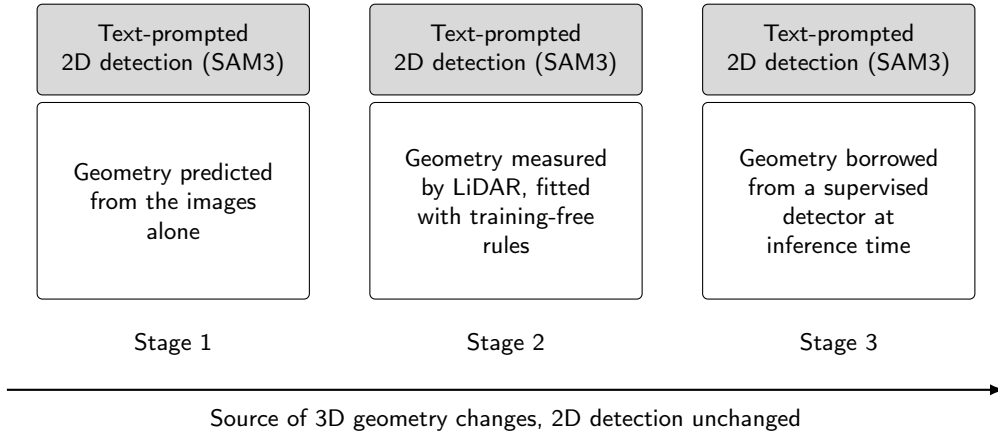
The dominant way to build such a detector is supervised learning: deep networks trained on large corpora of human-annotated 3D boxes [7, 22]. This approach works remarkably well, but it carries two structural costs. The first is annotation. Labeling sequences from a LiDAR (Light Detection and Ranging) sensor with oriented 3D boxes is among the most expensive labeling tasks in computer vision, and the effort does not transfer: a new sensor arrangement, a new city or a revised class list restarts it.

The second cost is subtler: a *closed vocabulary*. A detector trained on ten classes does not merely perform worse on an eleventh; it is structurally incapable of reporting it. Traffic, however, is not limited to ten categories: ambulances and fire trucks behave unlike ordinary trucks, a child is not an average pedestrian, and road debris belongs to no class at all. The class list of the benchmark used in this paper, nuScenes [1], excludes police vehicles, so a system that places a visually correct box on one is *penalized* for that detection. Extending a closed-vocabulary detector to any of these concepts means new labels and retraining.

Meanwhile, image understanding has moved in the opposite direction. Foundation models trained on web-scale image-text data recognize concepts that are named in free-form text rather than fixed in advance, an ability referred to as an *open vocabulary*. Promptable segmentation models such as SAM3 [2] take this one step further: given a text prompt, they return a mask for every instance of the named concept in the image. These models, however, return pixels, not positions, while driving requires metric 3D. The missing element is not geometry itself, since the vehicle already measures its surroundings with LiDAR, but the link between a named concept in the image and a metric box in the world.

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**Figure 1:** The three-stage comparison. The 2D detection is the same in all three stages, and only the source of the 3D geometry changes. The shaded box marks the component that is held fixed.

This paper asks how far that gap can be closed without any 3D annotation. The system it studies uses only text prompts, a promptable segmenter, and raw sensor geometry. Can such a system produce useful 3D detections, and what exactly separates it from supervised detectors?

The central idea is a three-stage comparison. Each stage is a complete version of the pipeline, evaluated on the nuScenes benchmark under its official protocol. Text-prompted 2D detection is held fixed across the stages, and only the source of the metric 3D geometry changes: in the first stage it is predicted from the images alone, in the second it is measured by LiDAR and fitted with training-free rules, and in the third it is borrowed at inference time from a supervised detector. Because a single factor changes at a time, the difference between two consecutive stages isolates the value of that factor, which separates the value of range sensing, of geometric fitting and of supervised geometry from the value of open-vocabulary detection. Figure 1 shows the three stages side by side; the shaded box is identical in every stage, and only the geometry below it changes.

In the final step the direction is reversed. The same masks are attached to a supervised LiDAR-only detector as a training-free *camera witness*: for every box the detector reports, the system checks whether the cameras contain a mask of the same class at the same place, and the confidence of the box is kept or lowered accordingly. This configuration is compared against a detector that learns its camera fusion from full supervision.

The main contributions of this paper are the following.

1. A training-free, open-vocabulary 3D detection pipeline that turns promptable segmentation masks into 3D boxes with simple, class-aware rules applied to the LiDAR points, evaluated end-to-end under the official nuScenes protocol and accompanied by a study of prompt design and by an inference procedure that makes evaluation on the full validation split possible on a single GPU.
2. A camera-only variant of the same pipeline, which separates the contribution of measured range from that of open-vocabulary detection.
3. A method for attaching supervised box geometry to the pipeline’s own detections at inference time, which shows that the largest deficit of the pipeline is measurement precision rather than 2D detection, while two further limits, class confusion and confidence calibration, survive that substitution.
4. A training-free camera witness that improves a supervised LiDAR-only detector, recovering roughly half of the gain of fully supervised camera fusion at zero annotation cost.
5. A coverage analysis of open-vocabulary 3D detection: which objects are found, which are misnamed, and where coverage ends.

## 2. Related work

### 2.1. Supervised 3D object detection

A LiDAR-based 3D detector takes as input a point cloud, the set of 3D points measured by the spinning laser scanner, and outputs a list of oriented 3D boxes, each with a class label and a confidence score. Modern benchmarks—KITTI [4] and, at larger scale, nuScenes [1]—established the now-standard formulation of this task: oriented boxes over a fixed taxonomy, learned from dense human annotation. VoxelNet [25] introduced end-to-end learning on voxelized point clouds, SECOND [21] made the voxel backbone practical with sparse convolutions, and pillar-based encoders [7] made bird’s-eye-view (top-down) processing of the point cloud efficient. Center-based designs such as CenterPoint [22] became the dominant paradigm on nuScenes: objects are detected as center points in a bird’s-eye-view map, and box dimensions, orientation and velocity are read off from the features at each center. CenterPoint plays two roles in this paper: it is the source of supervised geometry in the third stage, and the LiDAR-only detector to which the camera witness is attached.

Supervised detection also has a camera-only line. FCOS3D [19] predicts 3D boxes from a single image with a fully convolutional head, while DETR3D [20] and BEVFormer [9] aggregate multi-camera evidence through learned queries and bird’s-eye-view representations. These detectors use the same input as the camera-only variant studied here, but learn a closed vocabulary from full 3D supervision, whereas the variant here is training-free and open-vocabulary.

Multi-modal extensions inject camera information into the LiDAR pipeline. PointPainting [17] decorates each LiDAR point with semantic scores predicted from the images. MVP [23] densifies the point cloud with camera-derived *virtual points*, a design decision that matters for this paper: it means that even “LiDAR” detectors internally exploit dense image evidence. BEVFusion [11] fuses the two modalities in a shared bird’s-eye-view representation. All of these systems share the two costs that motivate this study: full 3D supervision and a closed vocabulary. The final part of this paper asks what a camera channel is worth to a LiDAR detector from the opposite end of the supervision spectrum, attaching the camera evidence at inference time through open-vocabulary masks and a fixed three-state rule, with no fusion training at all.

### 2.2. The nuScenes benchmark and evaluation protocol

Collected in Boston and Singapore, nuScenes [1] is a large-scale autonomous driving dataset. It contains 1,000 driving scenes of 20 seconds each, recorded with six cameras covering the full horizon, a 32-beam LiDAR spinning at 20 Hz, and five radars. Objects are annotated with oriented 3D boxes over ten classes on *keyframes*, the two frames per second at which all sensors are synchronized and labeled, and every object carries a visibility level recording how much of it is seen from the cameras. The validation split used throughout this paper contains 150 scenes with 6,019 keyframes and roughly 122,000 in-range annotated objects.

Under the official protocol, a predicted box matches a ground-truth object of the same class if their centers are closer than a threshold on the ground plane. Average precision (AP) is computed at four thresholds, 0.5, 1, 2 and 4 m, and averaged over thresholds and classes into mean average precision (mAP). Matching by center distance rather than by box overlap decouples *finding* an object from estimating its exact extent, which sparse LiDAR often cannot support. The quality of the matched boxes is measured separately by five true-positive errors: the average translation error (ATE, in meters), the average scale error (ASE), the average orientation error (AOE), the average velocity error (AVE) and the average attribute error (AAE). The nuScenes Detection Score (NDS) combines everything with mAP carrying half the weight,

$$\text{NDS} = \frac{1}{10} \left[ 5 \text{mAP} + \sum_{\text{mTP}} (1 - \min(1, \text{mTP})) \right], \quad (1)$$

so that a system is rewarded both for finding objects and for measuring them precisely. Each class is evaluated only within a class-specific range: 50 m for vehicles, 40 m for pedestrians and two-wheelers, and 30 m for traffic cones and barriers. Annotated objects with no LiDAR or radar return are excluded from evaluation.

### 2.3. Promptable segmentation and open-vocabulary perception

CLIP [14] demonstrated that image–text pretraining yields transferable open-vocabulary recognition. One line of work brought this ability to 2D detection. ViLD [5] distills CLIP knowledge into a detector, Detic [24] widens the vocabulary with image-level labels, and GLIP [8] and Grounding DINO [10] treat detection as the grounding of free-form text phrases. The Segment Anything family took the complementary segmentation-and-prompt side. The original Segment Anything Model (SAM) [6] introduced class-agnostic segmentation from geometric prompts: a click

or a box yields a mask for whatever object lies there, without naming its category. SAM2 [16] extended the same interface to video. SAM3 [2] added the capability this paper depends on: *concept prompts*. Given a short noun phrase, the model returns *all* instances of that concept in the image, each with its own mask and confidence score, which turns a segmentation model into an open-vocabulary detector whose vocabulary is decided at inference time. Its strength is that the class list becomes a list of words; its cost is that inference scales with the number of queried concepts, since each prompt is a separate query against the image. The pipeline developed here uses SAM3 as its sole 2D detection component: all category knowledge enters the system as text prompts, and everything downstream of the mask is training-free geometry.

## 2.4. Open-vocabulary and weakly supervised 3D detection

A growing body of work transfers 2D vision–language knowledge to 3D, along two main routes. The first *distills*: image–language features are projected into the point cloud and a 3D network is trained to reproduce them, as in OpenScene [13] and CLIP2Scene [3]. The second generates *pseudo-labels*: 2D detections are lifted to 3D boxes and used as training targets for a 3D detector, as in OV-3DET [12]. Both routes still train a 3D network on the target domain, although the pseudo-label generation step is close in spirit to the pipeline studied here. This paper differs in two respects: the entire system is inference-time only, with no 3D network trained at any stage, and it provides a complete, official-protocol evaluation together with a controlled decomposition that locates precisely what separates such training-free systems from supervised ones.

## 2.5. Camera-only metric geometry

Estimating metric geometry from images alone is a long-standing problem. Classical multi-view stereo requires overlapping viewpoints, and learned monocular depth was for years tied to the dataset it was trained on, with unreliable absolute scale. MiDaS [15] showed that training across mixed datasets yields robust relative depth, and DUST3R [18] established the *point-map* paradigm, a metric 3D coordinate for every pixel predicted directly from images, on which recent visual geometry transformers build. The Driving Visual Geometry Transformer [26] produces metric point maps from surround-view driving footage. It serves as the camera-only geometry backbone of the first stage: it replaces the LiDAR cloud under an otherwise identical pipeline, which isolates the contribution of true range sensing.

# 3. Method

The method is organized as a three-stage comparison. The 2D detection component is fixed throughout: an open-vocabulary, promptable segmentation model (SAM3) that localizes text-described concepts in the vehicle’s six surround-view cameras (Section 3.1). What changes from stage to stage is where the metric 3D geometry comes from: learned monocular geometry only (Section 3.2), raw LiDAR returns with training-free geometric fitting (Section 3.3), or the box geometry of a supervised LiDAR detector, associated to the pipeline’s detections at inference time (Section 3.4). Finally, the direction is reversed: the same open-vocabulary machinery is attached to the supervised detector as a training-free *camera witness* (Section 3.5). No component at any stage is trained or fine-tuned on nuScenes; the only domain-specific inputs are the class names and the sensor calibration and poses that the dataset provides. Section 3.6 collects the numeric values behind the design choices and states where each comes from.

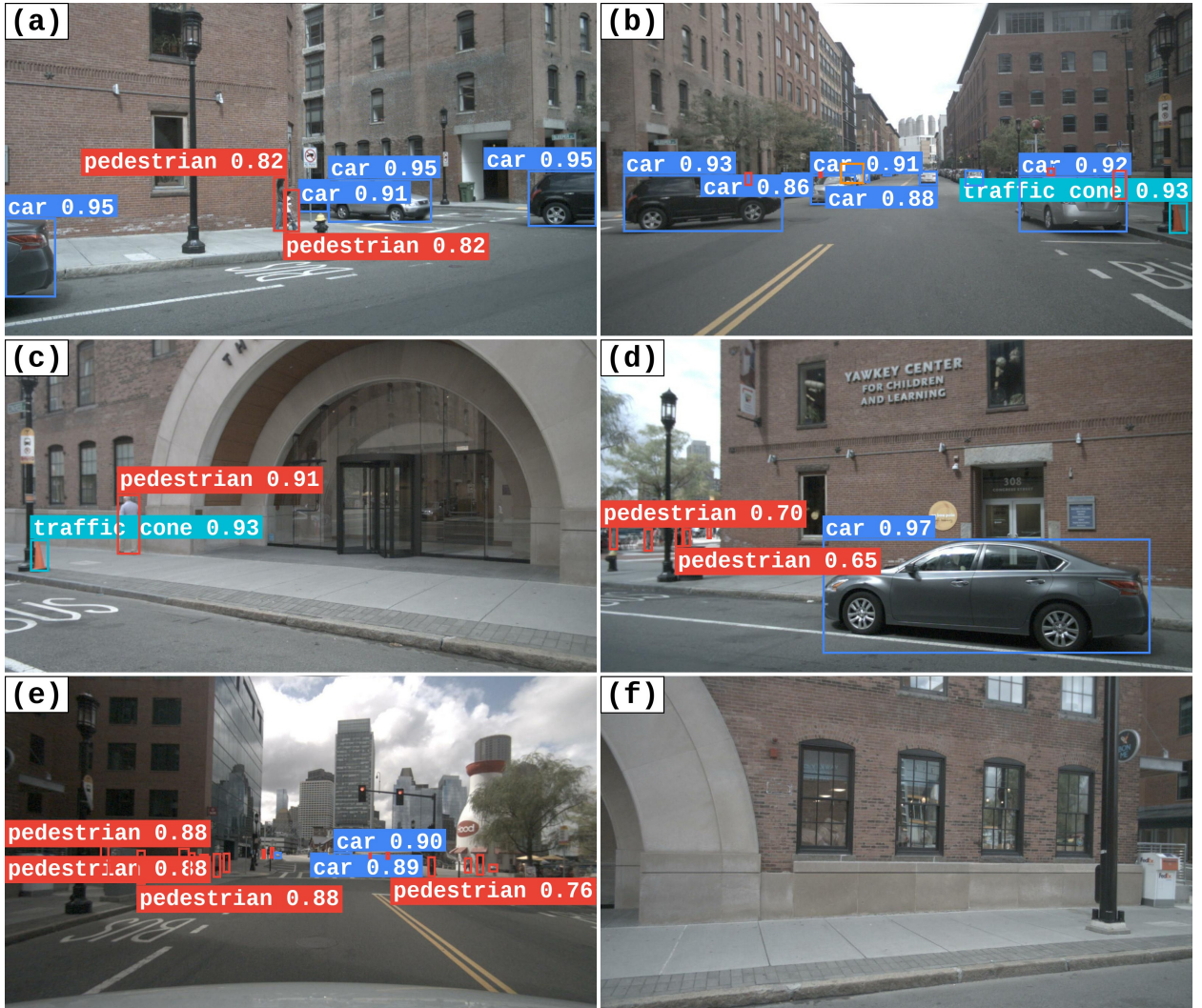
## 3.1. Promptable segmentation and prompt design

### 3.1.1. SAM3 as the 2D detection component

SAM3 [2] performs *promptable concept segmentation*: given a free-text prompt, it returns an exhaustive set of instance masks for that concept, each with a confidence score, without any task-specific training. This single capability supplies everything the pipeline needs from the images: knowledge of the category, separation of one object from another, and localization in 2D.

Figure 2 shows SAM3 detections on the six cameras of one validation frame. For readability the figure draws the bounding box of each mask rather than the mask itself, labeled with the predicted class and score; only detections scoring at least 0.5 are drawn, and repeated detections of the same object are shown once, while the pipeline itself uses every mask above the threshold described below.

A mask is not rejected on the strength of its score. Every mask that clears the size and evidence tests of Section 3.3 produces a box, whatever its confidence, and the ranking is left to the evaluation. The score of 0.2 in Table 3 is a separate matter: it is the confidence a detection must reach before the temporal layer of Section 3.3.2 uses it as evidence, as a



**Figure 2:** SAM3 detections on the six surround-view cameras of a single nuScenes keyframe [1]: (a) front left, (b) front, (c) front right, (d) back left, (e) back, (f) back right. Each mask is drawn as its bounding box, labeled with the prompt that produced it and with its score where the label fits. Only detections scoring at least 0.5 are drawn, while the pipeline itself uses every mask above the lower threshold.

node of a track or as a neighbor from which a velocity is read. A detection below it is still emitted and can still receive a velocity and an orientation from more confident neighbors; it simply cannot anchor others.

Keeping the emission threshold low is worth measuring. On the full validation split, lowering it from 0.5 to 0.2 roughly doubles the number of boxes, from 143,012 to 277,291, and adds 0.031 mAP and 0.013 NDS. The gain concentrates in the small classes, which the segmentation scores least confidently: pedestrians gain 0.067 AP, motorcycles 0.063, traffic cones 0.052 and bicycles 0.037, while cars gain 0.010 and trailers 0.001. Section 4.4 sweeps the threshold further. A permissive list is safe because a low-scoring mask must still survive the depth gate and the minimum-evidence test: masks that correspond to nothing measurable are removed by the geometry stage rather than by the score.

### 3.1.2. Prompt set

Thirteen prompts are queried per image: one per nuScenes detection class, plus three sub-class variants (Table 1). The variants matter. The generic prompt “construction vehicle” is a weak description of the excavators and bulldozers that dominate the class, and querying those concepts directly yields measurably better detections; “flatbed trailer” plays

**Table 1**

Text prompts used by the pipeline. The ten base prompts name the evaluated classes. The three variant prompts describe subtypes and are mapped onto the class they belong to.

| Role         | Prompts                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------|
| Base (10)    | car, truck, bus, pedestrian, bicycle, motorcycle, traffic cone, barrier, trailer, construction vehicle |
| Variants (3) | excavator, bulldozer, flatbed trailer                                                                  |

the same role for trailers. Detections produced by a variant prompt are reported under their parent class: an “excavator” mask becomes a construction-vehicle detection.

Prompt wording is not cosmetic. An early version of the pipeline used the prompt “person”, which turned out to segment *depictions* of people, such as figures on advertising billboards, alongside actual road users; switching to “pedestrian” aligned the concept with the intended traffic participant. The choice of prompt thus acts as a zero-cost prior on what the system considers an object, and prompt engineering replaces what would otherwise require retraining in a closed-vocabulary detector.

### 3.1.3. Inference cost

The flexibility of text prompts has a price: inference cost grows with the number of queried concepts, since each prompt is a separate query against the image. This cost is reduced in three ways: each image is encoded once and all thirteen prompts run against that shared encoding, the six cameras of a keyframe are processed as a single batch, and data preparation and box fitting are overlapped with GPU inference. Together these measures make the pipeline roughly fifteen times faster than the initial implementation on equal hardware, so that a full run over the 150-scene validation split completes in about one hour on a single GPU, with no measurable change in accuracy.

## 3.2. Camera-only lifting with learned monocular geometry

The first stage uses no range sensor. *Lifting* means turning a 2D mask in the image into a 3D box in the world, and each stage is named after the source of the 3D information it lifts with. The Driving Visual Geometry Transformer (DVGT-2) [26] predicts a metric *point map*—a 3D coordinate per pixel—from surround-view footage. Each SAM3 mask is lifted by reading the predicted 3D points at its pixels and taking their median as the object’s center, and the box is completed with class-prior dimensions, the typical size of the class. This variant is fully camera-only: both the detections and the geometry come from images. Everything downstream of the lifted center, namely the class-prior dimensions, duplicate removal and the temporal layer, is shared with the LiDAR-based variant, so the two stages differ only in where their geometry comes from.

Its failure mode is instructive. Monocular geometry is *inferred* rather than measured, since metric scale has to be read from appearance alone, so range errors grow with distance and with object size, and the predicted surface can drift smoothly with no local cue that it has done so. The consequence, quantified in Section 4, is that camera-only lifting finds objects well but cannot place them precisely enough for the strict center-distance thresholds of the 3D metric.

The two components meet on the image grid. SAM3 produces its masks at the full camera resolution of 1600×900, while the point map is defined on the model’s own 512×288 grid, so a mask is downsampled to that grid before the points under it are read. DVGT-2 returns the map in the ego frame of the first keyframe of the scene, whose pose carries the lifted centers into the global frame of the evaluation.

## 3.3. LiDAR lifting with a depth gate

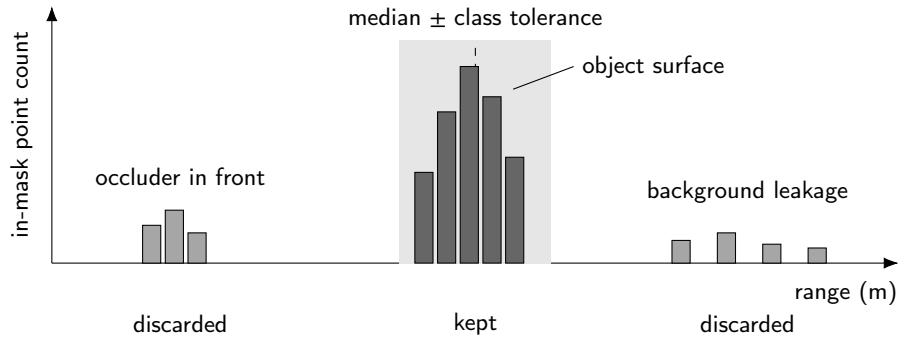
The second stage keeps the same masks and class names, and replaces predicted geometry with measured geometry. The single-sweep LiDAR cloud of the keyframe, the points from one full rotation of the scanner, is projected into each camera, and the points that fall inside a mask are assigned to that mask. Two details make this association reliable:

- **Depth gate.** A 2D mask is depth-blind: a mask over a distant object also covers foreground points on occluders, and points leaking through mask boundaries land on the background. Assigned points whose range deviates from the in-mask median by more than a class-dependent tolerance (Table 2) are discarded.
- **Minimum evidence.** A mask must cover at least the class-dependent pixel count of Table 2, and at least three of its assigned points must survive the depth gate, before it produces a detection; this suppresses masks that see an object the LiDAR cannot confirm.

**Table 2**

Per-class pipeline parameters. The depth-gate tolerance scales with the physical length of the class. The range limits reproduce the class-specific ranges of the official evaluation protocol. Classes whose center is taken from the point median use no box fit and therefore no fit threshold.

| Class        | Min. mask (px) | Depth gate (m) | Range (m) | Center from | Fit thr. (m) | Prior $L \times W \times H$ (m) |
|--------------|----------------|----------------|-----------|-------------|--------------|---------------------------------|
| Car          | 244            | 2.5            | 50        | box fit     | 2.4          | $4.62 \times 1.95 \times 1.73$  |
| Truck        | 244            | 4.0            | 50        | box fit     | 3.0          | $6.93 \times 2.51 \times 2.84$  |
| Bus          | 244            | 6.0            | 50        | box fit     | 4.5          | $11.00 \times 2.94 \times 3.47$ |
| Trailer      | 244            | 6.0            | 50        | box fit     | 4.0          | $12.29 \times 2.90 \times 3.87$ |
| Constr. veh. | 244            | 4.0            | 50        | box fit     | 3.0          | $6.37 \times 2.85 \times 3.19$  |
| Pedestrian   | 100            | 1.5            | 40        | median      | —            | $0.73 \times 0.67 \times 1.77$  |
| Motorcycle   | 100            | 1.5            | 40        | median      | —            | $2.11 \times 0.77 \times 1.47$  |
| Bicycle      | 100            | 1.5            | 40        | median      | —            | $1.70 \times 0.60 \times 1.28$  |
| Traffic cone | 50             | 1.0            | 30        | median      | —            | $0.41 \times 0.41 \times 1.07$  |
| Barrier      | 100            | —              | 30        | segments    | —            | $0.50 \times 2.53 \times 0.98$  |



**Figure 3:** The depth gate. The ranges of the points assigned to one mask separate into an occluder in front, the surface of the object itself, and background leaking through the mask boundary. Points outside a band around the median range are discarded. The distribution is drawn schematically.

Both the depth-gate tolerance and the fit threshold are derived from the class prior: the tolerance from the length of the class, with a floor for the small classes, and the fit threshold at a fixed fraction of the same length. Points on a 12-meter trailer legitimately spread over several meters of range, while a traffic cone occupies half a meter, so a single tolerance would either leak occluders into small objects or truncate large ones. The range limits reproduce the class-specific ranges of the official protocol. Barriers are not gated but segmented along their principal axis, as described below. Figure 3 illustrates the gate.

The critical property of this stage is that distance is now *measured*: the position error that remains comes from which points of the object the mask happened to capture, not from scale drift.

### 3.3.1. Heuristic box fitting

The gated points are reduced to an oriented box by deliberately simple, class-aware, training-free rules:

- **Large vehicles** (car, truck, bus, trailer, construction vehicle): the gated points are projected onto the ground plane and a rectangle is fitted to them. Yaw is searched in 30 steps of  $3^\circ$  over a quarter turn; for each hypothesis a rectangle is spanned by the 2nd and 98th percentiles of the rotated points, scored by the mean squared distance of the points to the nearest edge, and the lowest score wins. Since LiDAR points concentrate on the faces visible to the sensor, the rectangle aligns with those faces rather than with the full object. The long axis is read from the observed extent when that extent exceeds the class threshold of Table 2, and from the viewing direction otherwise, since a vehicle seen head-on shows only its short side. Along each axis the center is the midpoint of the observed extent when that extent reaches the prior dimension; otherwise, when the observed interval lies entirely on one side of the sensor, the observed edge nearer the sensor is kept and the prior dimension is extended

away from the sensor, and in the remaining case the midpoint is used. The fit requires at least eight gated points; below that, the center is the point median and the yaw is set to zero and left to the temporal layer.

- **Small objects** (pedestrian, bicycle, motorcycle, traffic cone): the point median is taken as the center; extent estimation from partial masks is unreliable at this scale.
- **Barriers**: the point set is segmented into 2.5 m pieces along its principal axis. A piece is emitted only if it holds at least three points, and its center is the median of those points. This matches the annotation convention of nuScenes.

Every detection inherits the score of its source mask, and the box's vertical position is set from the median height of the retained points.

### 3.3.2. Temporal post-processing

A light rule-based temporal layer supplies what a single frame cannot. Detections are first deduplicated per class by center distance, since the same object can be segmented in two overlapping cameras or by two prompts; the radius is larger for vehicles than for small objects, because two pedestrians can stand closer together than two cars can park. A greedy tracker then links detections across keyframes: each track is extended by the nearest unclaimed detection of the same class within a radius that grows with the number of keyframes since the track was last seen, and a track survives a gap of at most two keyframes.

Velocity is estimated from the displacement between symmetric neighbor keyframes, half a second apart, and is accepted only when the two half-steps agree, either both negligible or both in the same direction. Small magnitudes are then set to zero and large ones capped, which suppresses spurious velocities caused by association errors. Traffic cones and barriers are stationary by construction.

Orientation is resolved by a cascade of three rules: the direction of the track, when the track shows genuine displacement; otherwise the velocity direction; and for parked vehicles the driving side of the scene's city, which settles the remaining 180° ambiguity, since a parked car almost always faces the direction of traffic on its side of the road.

### 3.4. Bridging to supervised geometry

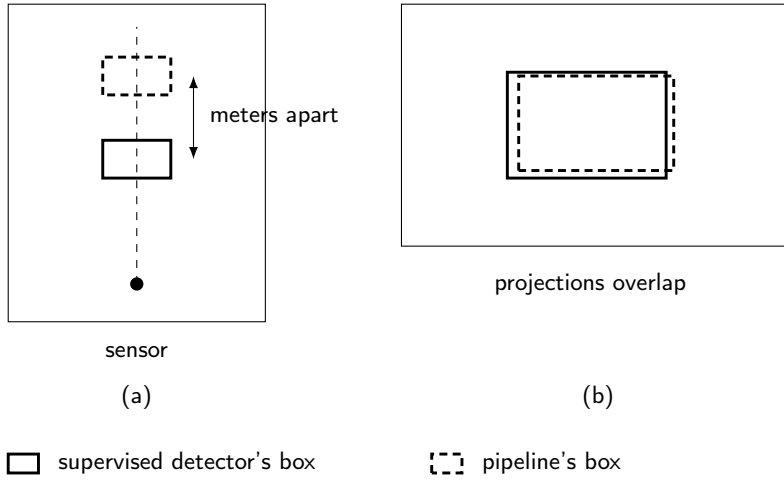
The heuristic fit of Section 3.3 improves on camera-only lifting, but it remains bounded by partial visibility: the mask sees one face of the object, the LiDAR samples that face sparsely, and the fitted center is pulled toward the visible surface. The third stage measures how much of the remaining deficit is geometry rather than 2D detection. It keeps the pipeline's detections, with classes and scores untouched, and replaces only their geometry with that of a supervised LiDAR detector, CenterPoint [22], associated at inference time. This stage is a diagnostic construction rather than a deployment proposal.

Association proceeds in two channels, in order, both greedy and one-to-one: detections are processed in descending score order, and every CenterPoint box can be claimed by at most one detection.

1. **Ground-plane proximity.** Each pipeline detection claims the nearest same-class CenterPoint box within 2 m in the bird's-eye view.
2. **Image-space association.** Detections left unmatched are re-tried in the image. The pipeline's range errors are predominantly *radial*, along the line of sight from the sensor, and a radial error barely moves a projection, so the same physical object yields overlapping projections even when the two boxes are meters apart in the ground plane. A detection claims a same-class CenterPoint box whose projections overlap in any camera, with a containment relaxation for the case where one projection sits entirely inside the other.

Neither threshold is fitted. The 2 m radius is one of the four matching distances of the official evaluation protocol, and the image-space channel accepts an overlap of at least 0.25 intersection-over-union. Figure 4 shows why the second channel works: the two boxes of the same object lie meters apart along the viewing ray, yet their projections in the camera almost coincide.

Matched detections adopt the CenterPoint box together with its orientation and velocity; unmatched detections keep their heuristic geometry. The second channel raises the matched share precisely on the distant objects whose heuristic fit is worst.



**Figure 4:** Radial error and image-space association. (a) In the ground plane, seen from above, two boxes for the same object sit meters apart along the line of sight from the sensor. (b) In the camera, their projections still overlap. The solid box is the supervised one and the dashed box is the pipeline's.

### 3.5. A training-free camera witness in the reverse direction

The final stage reverses the roles. CenterPoint is a LiDAR-only detector, and the same open-vocabulary machinery is attached to it as a training-free camera channel. Every CenterPoint box is projected into the cameras and tested against the SAM3 masks of the same keyframe, using an image overlap of at least 0.25 intersection-over-union; three variants of this test differ by at most 0.003 mAP (Section 4.7). The outcome is one of three evidence states, each mapped to a fixed score multiplier. A box is *confirmed* when a same-class mask overlaps its projection; its score is kept. It is *contradicted* when masks overlap but all carry a different class; its score is multiplied by 0.85. It is *unwitnessed* when no mask overlaps; its score is multiplied by 0.6. The intuition is that of a second witness: a real object should normally be visible to the cameras, so a box that no mask corroborates is more often a phantom, while a box the masks contradict may well have the right geometry with only the name in dispute. No box is moved, resized or removed; only scores change. The rule is deliberately minimal, with three states, two hand-set multipliers that are not the best of the six settings evaluated in Section 4.7, and no training, so any gain is attributable to the open-vocabulary evidence itself rather than to a fitted model. Figure 5 summarizes the rule.

Section 4.7 compares this channel against MVP [23], which learns its camera fusion from full supervision.

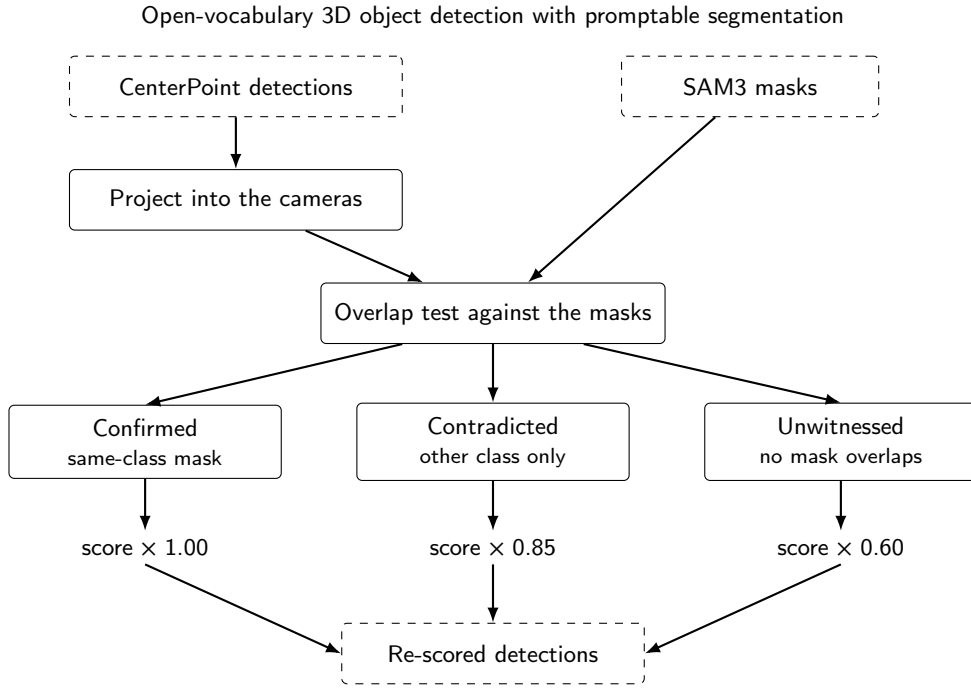
### 3.6. Parameters

The numeric values behind the design choices of this section are collected in Table 3, and the class-dependent ones in Table 2. The last column of Table 3 says where each value comes from. *Protocol* and *dataset* mark values fixed by the official evaluation protocol or by nuScenes itself; *swept* marks values measured over a range, and *varied* marks the witness overlap test, for which three forms are compared. Everything else is an engineering default, held fixed across every experiment in this paper. The class priors of Table 2 are the mean box dimensions of each class on the training split of nuScenes, not on the validation split: every value agrees with the training mean to within 0.2 m, and eight of the ten to within 0.05 m.

## 4. Results

### 4.1. Experimental setup

All experiments are conducted on the official nuScenes validation split of 150 scenes (6,019 keyframes) and evaluated with the official *DetectionEval* protocol of Section 2.2: mAP over four center-distance thresholds, the five true-positive error measures, and NDS as the combined score, with class-specific range limits and ground-truth filters applied exactly as in the official implementation. After filtering, the split contains 121,861 ground-truth objects, dominated by cars (53,043) and pedestrians (23,500). The protocol scores exactly these ten classes, so the open-vocabulary ability of the pipeline is measured only on this fixed list, and detections outside it count as false positives.



**Figure 5:** The camera-witness rule. A supervised box is projected into the cameras and tested against the masks of the same keyframe. Its score is kept when a mask of the same class overlaps it, reduced when the overlapping masks all carry a different class, and reduced further when no mask overlaps at all.

**Table 3**

Parameters of the pipeline. None of these values is fitted on the evaluation split, and the last column says where each one comes from. The class-dependent quantities are listed separately in Table 2.

| Parameter                                              | Stage          | Value                                               | Basis    |
|--------------------------------------------------------|----------------|-----------------------------------------------------|----------|
| Mask score threshold for temporal evidence             | Temporal layer | 0.2                                                 | Default  |
| Minimum gated points per mask                          | LiDAR lifting  | 3; 8 for the rectangle fit                          | Default  |
| Barrier segment length                                 | LiDAR lifting  | 2.5 m                                               | Dataset  |
| Duplicate-removal radius                               | Temporal layer | 2 m for vehicles, 1 m for barriers, 0.5 m otherwise | Default  |
| Track association radius                               | Temporal layer | 10 m per keyframe of gap                            | Default  |
| Largest gap a track survives                           | Temporal layer | 2 keyframes                                         | Default  |
| Keyframe spacing                                       | Temporal layer | 0.5 s                                               | Dataset  |
| Neighbor distance accepted for velocity                | Temporal layer | 10 m                                                | Default  |
| Velocity clamp                                         | Temporal layer | below 1 m/s set to zero, above 15 m/s capped        | Default  |
| Track displacement before orientation is taken from it | Temporal layer | 3 m and 1 m/s                                       | Default  |
| Ground-plane association radius                        | Bridge         | 2 m                                                 | Protocol |
| Image-space association threshold                      | Bridge         | 0.25 IoU                                            | Default  |
| Witness overlap threshold                              | Witness        | 0.25 IoU                                            | Varied   |
| Witness score multipliers                              | Witness        | 0.85 contradicted, 0.60 unwitnessed                 | Swept    |

The compared systems are the three stages of Section 3, together with the two supervised detectors that serve as references:

- **SAM3 + DVG2-2:** open-vocabulary masks lifted to metric 3D with the Driving Visual Geometry Transformer (Section 3.2);
- **SAM3 + LiDAR:** the training-free pipeline of Section 3.3;

- **SAM3 + CenterPoint geometry:** the same detections with supervised box geometry associated at inference time (Section 3.4);
- **CenterPoint + SAM3 witness:** the supervised LiDAR detector re-scored by the camera witness rule (Section 3.5);
- **CenterPoint and MVP:** the supervised reference detectors, LiDAR-only and LiDAR+camera respectively.

Detection in the images uses the SAM3 image model with the official checkpoint. Each 1600×900 camera image is resized to 1008×1008 for the network, and the predicted masks are mapped back to the original size. Masks are binarized at a probability of 0.5, and the model’s own detection threshold is 0.05. Inference runs in bfloat16 on a single NVIDIA A100 GPU, six camera images at a time. The validation split is processed once and the resulting masks, boxes and in-mask point indices are stored, so that every later experiment reuses the same 2D detections. Camera-only geometry uses DVGT-2 with its official checkpoint and DINOv3 image backbone in causal streaming mode, one frame at a time, with the trajectory head disabled. The supervised references are the publicly released validation predictions of CenterPoint and MVP; their scores in Table 5 reproduce the published validation figures, and neither detector is retrained or given access to the pipeline’s outputs. Table 4 pins every component to the revision that produced the results of this section.

## 4.2. Main results of the three-stage comparison

Table 5 reports the main comparison, ordered by the source of 3D geometry.

Each step isolates one question. *Camera-only to LiDAR* (0.183 → 0.298 mAP): measured range is worth +0.115 mAP over learned monocular range under an otherwise identical pipeline, the value of the sensor separated from the value of open-vocabulary detection. *Heuristic to supervised geometry* (0.298 → 0.413 mAP, 0.348 → 0.555 NDS): keeping the pipeline’s detections and adopting CenterPoint boxes where a match exists lifts NDS by +0.21. The largest deficit of the training-free pipeline therefore lies in measurement precision, in position, extent, orientation and velocity, rather than in 2D detection; Section 4.6 shows that it is not the only one. *LiDAR-only to LiDAR+camera* (lower block and fourth row): a camera channel helps a LiDAR detector in both regimes. MVP learns it from full supervision and gains +0.074 mAP over CenterPoint; the training-free witness rule gains +0.034 mAP, roughly half of the trained-fusion gain.

The five right-hand columns of Table 5 break the true-positive errors down across the same configurations: mATE is a distance in meters, mAVE a speed in meters per second, mAOE an angle in radians, and mASE and mAAE are dimensionless. Replacing the heuristic fit by supervised boxes shrinks every term at once: translation falls from 0.634 to 0.379, orientation from 0.967 to 0.385 and velocity from 0.733 to 0.297. Scale improves least, from 0.347 to 0.266, as expected for a pipeline that takes its extents from a class prior. What remains between the third row and CenterPoint itself is almost entirely translation, 0.379 against 0.292. The witness rule moves no box, so its five terms are those of CenterPoint to within 0.003, and its gain appears in mAP and NDS alone.

Table 6 gives the per-class breakdown of the training-free pipeline.

## 4.3. The temporal layer

Table 7 isolates the temporal layer of Section 3.3.2. Every row uses the same detections and the same boxes, so mAP stays at 0.298, mATE at 0.634 and mASE at 0.347 throughout, and the layer’s entire contribution appears in the three remaining error terms and in NDS. Removing it costs 0.045 NDS, and the path between the two ends is not monotone.

A property of the metric explains the shape of the table: NDS credits a true-positive error term only when it falls below 1.0, so resolving orientation in the third row improves mAOE from 1.311 to 1.099 without moving NDS, and the gain is collected only in the fifth row, where the velocity channel crosses below 1.0. Two rows are informative failures. Narrowing the velocity radius zeroes the velocity of genuinely moving objects and drags the attribute error with it. Taking velocity from the track is worse than taking it from symmetric neighbor keyframes, because a track that jumps identity carries that jump into its velocity; taking only the orientation from the track is safe, and the eighth row combines the two. The velocity clamp is the smallest component, worth 0.003 NDS.

**Table 4**

Model and reference versions. Weights are identified by their Hugging Face repository and code by its GitHub repository, and a version is the first seven characters of the commit identifier. The two supervised references are identified by the configuration name of their released model.

| Component               | Source                  | Version                                      |
|-------------------------|-------------------------|----------------------------------------------|
| SAM3 weights            | facebook/sam3           | 3c879f3                                      |
| SAM3 code               | facebookresearch/sam3   | 46957e4                                      |
| DVGT-2 weights          | RainyNight/DVGT-2       | 48fcd2                                       |
| DVGT-2 code             | wzzheng/DVGT            | 51cf3f6                                      |
| DINOv3 backbone         | facebookresearch/dinov3 | 6876159                                      |
| CenterPoint predictions | tianweiy/CenterPoint    | nusc_centerpoint_voxelnet_0075voxel_fix_bn_z |
| MVP predictions         | tianweiy/MVP            | nusc_two_stage_base_with_virtual             |

**Table 5**

Main results on the nuScenes validation split, ordered by the source of 3D geometry. The upper block is training-free and open-vocabulary in its 2D detection, and the lower block lists supervised references. The input column gives C for camera and L for LiDAR, and lower is better in the five right-hand columns.

| Method                      | Input | mAP   | NDS   | mATE  | mASE  | mAOE  | mAVE  | mAAE  |
|-----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| SAM3 + DVGT-2               | C     | 0.183 | 0.282 | 0.691 | 0.484 | 0.843 | 0.738 | 0.342 |
| SAM3 + LiDAR                | C+L   | 0.298 | 0.348 | 0.634 | 0.347 | 0.967 | 0.733 | 0.331 |
| SAM3 + CenterPoint geometry | C+L   | 0.413 | 0.555 | 0.379 | 0.266 | 0.385 | 0.297 | 0.189 |
| CenterPoint + SAM3 witness  | C+L   | 0.630 | 0.685 | 0.295 | 0.255 | 0.304 | 0.257 | 0.191 |
| CenterPoint                 | L     | 0.596 | 0.668 | 0.292 | 0.255 | 0.302 | 0.259 | 0.194 |
| MVP                         | C+L   | 0.670 | 0.707 | 0.289 | 0.251 | 0.281 | 0.270 | 0.189 |

**Table 6**

Per-class results of the training-free pipeline on the validation split. The true-positive error measures of the same configuration are given in Table 5.

| Class        | AP    | Class        | AP    |
|--------------|-------|--------------|-------|
| Car          | 0.371 | Pedestrian   | 0.637 |
| Truck        | 0.220 | Motorcycle   | 0.454 |
| Bus          | 0.321 | Bicycle      | 0.240 |
| Trailer      | 0.009 | Traffic cone | 0.382 |
| Constr. veh. | 0.114 | Barrier      | 0.233 |

**Table 7**

Effect of the temporal layer. Every row uses the same detections and the same boxes, so mAP, mATE and mASE are identical throughout and only the three remaining error terms move. The three indented rows each modify the fifth, and the eighth is the configuration used throughout this paper. The final row removes the velocity clamp from that configuration.

| Configuration                                | mAOE  | mAVE  | mAAE  | NDS   |
|----------------------------------------------|-------|-------|-------|-------|
| No temporal layer, single frame              | 1.311 | 1.318 | 0.482 | 0.303 |
| Velocity                                     | 1.311 | 1.081 | 0.403 | 0.311 |
| Velocity and orientation                     | 1.099 | 1.081 | 0.403 | 0.311 |
| Narrow velocity radius with parked prior     | 1.200 | 1.259 | 0.452 | 0.306 |
| Wide velocity radius, selective parked prior | 1.006 | 0.733 | 0.331 | 0.345 |
| with the parked prior on two-wheelers        | 1.018 | 0.733 | 0.331 | 0.345 |
| with velocity from the track                 | 0.978 | 0.963 | 0.341 | 0.323 |
| with orientation from the track              | 0.967 | 0.733 | 0.331 | 0.348 |
| Final row without the velocity clamp         | 0.967 | 0.758 | 0.333 | 0.345 |

**Table 8**

Sensitivity to the score threshold. Each row is a separate evaluation of the same detections under the official protocol, keeping only those above the stated score. The result changes little across the range, so the value used is not a peak.

| Score threshold | mAP   | mATE  | mAOE  | mAVE  | NDS   |
|-----------------|-------|-------|-------|-------|-------|
| 0.05            | 0.298 | 0.634 | 0.967 | 0.733 | 0.348 |
| 0.10            | 0.295 | 0.628 | 0.957 | 0.713 | 0.350 |
| 0.15            | 0.290 | 0.625 | 0.954 | 0.701 | 0.350 |
| 0.20            | 0.285 | 0.624 | 0.952 | 0.695 | 0.348 |
| 0.25            | 0.280 | 0.623 | 0.951 | 0.690 | 0.347 |
| 0.30            | 0.276 | 0.622 | 0.949 | 0.687 | 0.345 |
| 0.40            | 0.267 | 0.620 | 0.949 | 0.683 | 0.341 |
| 0.50            | 0.254 | 0.620 | 0.948 | 0.680 | 0.335 |

#### 4.4. Sensitivity to the score threshold

The pipeline emits every detection whose mask score clears a low threshold and leaves the ranking to the evaluation. Table 8 sweeps that threshold, with a separate official evaluation at each value, to test whether the low value was chosen because it happens to be favorable.

The result is flat rather than peaked. Raising the threshold from 0.05 to 0.50 discards six of every seven detections and costs 0.044 mAP and 0.013 NDS. Even at 0.40 the pipeline emits 63,622 car detections against 53,043 annotated cars, so the sweep stays on the permissive side throughout. The true-positive errors improve slightly, because the detections discarded first are the weakest. AP is computed from a ranked list, so a long low-scoring tail is close to free: it fills the low-recall end of the curve without displacing anything above it.

#### 4.5. Camera-only comparison

The first two rows of Table 5 ask what a LiDAR adds to a camera system. Adding it is worth 0.115 mAP and 0.066 NDS, and since the 2D detections are identical, the whole difference is in placement. A box that names the object correctly but sits a meter and a half away matches at the two loose distances of the protocol and fails at the two tight ones, and the tight distances are demanding even with measured range, since the mean translation error of the LiDAR-based pipeline is 0.634 m. The camera-only row therefore separates the two halves of the problem: open-vocabulary detection transfers to 3D unchanged, and metric placement does not.

#### 4.6. Decoupling detection from geometry

The third stage defines what the open-vocabulary pipeline is worth when it no longer has to supply the geometry itself (Table 5, second and third rows); only the box geometry is exchanged.

The association channel matters. Matching on ground-plane proximity alone, within the 2 m radius, reaches mAP 0.351 / NDS 0.519: the pipeline's position errors at range routinely exceed 2 m, so exactly the detections that need supervised geometry most fail to receive it. Adding the image-space channel of Section 3.4, which is immune to radial error, raises the matched share from 16 % to 22 % of detections and the result to 0.413/0.555. Two controls complete the picture. First, the remaining gap to CenterPoint is not a matching artifact: relabeling CenterPoint's own boxes through 2D mask association plateaus at 0.37–0.40 mAP regardless of box quality, so the open-vocabulary *class* channel, dominated by truck–trailer and cone–barrier confusions, is the binding constraint. Second, scores from different sources cannot be mixed within one ranking: replacing scores for only the matched subset of the pipeline's boxes *degrades* mAP by 0.019, since the two score distributions are not on a common scale.

#### 4.7. Adding a camera channel to a LiDAR detector

The three-state witness rule of Section 3.5 lifts CenterPoint from 0.596 mAP / 0.668 NDS to 0.630 mAP / 0.685 NDS.

Two properties of this result stand out. First, the outcome barely depends on how the witness test is scored: whether a box and a mask are matched by projection overlap, by mask containment, or by a hybrid channel based on LiDAR points, the result changes by less than  $\pm 0.003$  mAP. What carries the signal is whether a same-class mask exists at all, and that is governed by visibility. Second, the comparison with MVP shows what supervision is worth: learned camera fusion gains +0.074 mAP, the training-free witness +0.034, so the witness recovers roughly half the benefit of supervised fusion at zero annotation cost.

**Table 9**

Witness score multipliers. Each row is a complete evaluation of the re-scored CenterPoint on the full validation split, and the row 1.00 / 0.85 / 0.60 is the setting used in this paper.

| Confirmed | Contradicted | Unwitnessed | mAP   | NDS   |
|-----------|--------------|-------------|-------|-------|
| 1.00      | 1.00         | 1.00        | 0.596 | 0.668 |
| 1.00      | 1.00         | 0.60        | 0.621 | 0.680 |
| 1.00      | 0.90         | 0.75        | 0.623 | 0.681 |
| 1.00      | 0.85         | 0.85        | 0.617 | 0.679 |
| 1.00      | 0.85         | 0.60        | 0.630 | 0.685 |
| 1.00      | 0.70         | 0.40        | 0.635 | 0.687 |

**Table 10**

SAM3 2D detection coverage of in-range ground-truth objects by class. The same-class column requires the overlapping mask to carry the correct name, the any-class column does not, and the misnamed column is the difference between the two. Classes are ordered by same-class coverage.

| Class        | GT      | Same class | Any class | Misnamed |
|--------------|---------|------------|-----------|----------|
| Bus          | 2,085   | 93.5 %     | 95.2 %    | 1.7 %    |
| Car          | 59,522  | 90.6 %     | 91.2 %    | 0.6 %    |
| Truck        | 9,764   | 86.8 %     | 92.3 %    | 5.5 %    |
| Barrier      | 16,911  | 85.2 %     | 86.7 %    | 1.5 %    |
| Trailer      | 2,446   | 79.5 %     | 89.2 %    | 9.7 %    |
| Pedestrian   | 25,678  | 77.1 %     | 79.5 %    | 2.4 %    |
| Traffic cone | 12,491  | 72.1 %     | 85.8 %    | 13.7 %   |
| Bicycle      | 2,060   | 72.1 %     | 79.2 %    | 7.1 %    |
| Motorcycle   | 2,009   | 70.1 %     | 82.8 %    | 12.7 %   |
| Constr. veh. | 1,599   | 57.5 %     | 76.7 %    | 19.2 %   |
| Total        | 134,565 | 84.2 %     | 87.5 %    | 3.3 %    |

The two multipliers matter as little as the overlap test. Table 9 evaluates six settings on the full split. Every setting that penalizes weak evidence improves on plain CenterPoint, and the spread across them is smaller than the gain itself. The values used throughout this paper, 0.85 and 0.60, were fixed before this sweep and are not the best of the six: penalizing more sharply, with 0.70 and 0.40, reaches 0.635 mAP and 0.687 NDS.

## 5. Discussion

This section dissects where the training-free pipeline succeeds and where its losses originate: what SAM3 detects in the image, how detection degrades with visibility and distance, and why detecting an object is not the same as scoring it.

### 5.1. 2D detection coverage by class

The first analysis asks whether, before any 3D reasoning, SAM3 finds the object in the image at all. For every ground-truth object within the class range limits, the test checks whether a mask of score at least 0.2 overlaps the object’s image projection at an intersection-over-union (IoU) of at least 0.25. The strict version accepts only a mask of the same class, the permissive version a mask of any class (Table 10), and the gap between the two isolates the objects that are segmented but misnamed. The population contains 134,565 objects, filtered only by the range limits; the official evaluation additionally discards annotations without any LiDAR or radar return, which gives the 121,861 objects of Section 4.

Zero-shot coverage is high where objects are large and visually distinctive (bus 93.5 %, car 90.6 %) and degrades for small or rare categories. More consequential, the gap between the two versions of the test isolates *misnaming*. For construction vehicles, one object in five is segmented but carries the wrong label, typically “truck”; for traffic cones the dominant competing label is “barrier”, and for motorcycles the two-wheeler prompts compete with each other. For these classes the limiting factor is not whether SAM3 sees the object but what it calls it—a prompt-taxonomy problem, which the sub-class prompts of Section 3.1 target directly.

**Table 11**

Box-level coverage by annotated visibility band and range. Each cell gives the number of ground-truth objects and the fraction of them the pipeline covers. Visibility is the annotated fraction of the object that is visible across the cameras.

| Visibility | 0–15 m        | 15–30 m       | 30 m+         | Total          |
|------------|---------------|---------------|---------------|----------------|
| 0–40 %     | 3,537 · 80 %  | 10,182 · 59 % | 9,749 · 33 %  | 23,468 · 52 %  |
| 40–60 %    | 1,915 · 92 %  | 5,783 · 79 %  | 5,091 · 54 %  | 12,789 · 71 %  |
| 60–80 %    | 3,713 · 96 %  | 7,964 · 85 %  | 6,154 · 60 %  | 17,831 · 79 %  |
| 80–100 %   | 24,135 · 96 % | 29,686 · 87 % | 14,054 · 68 % | 67,875 · 86 %  |
| Total      | 33,300 · 94 % | 53,615 · 81 % | 35,048 · 55 % | 121,963 · 77 % |

**Table 12**

2D detection coverage versus official AP for selected classes. The two columns do not move together, which separates being found in the image from scoring in the metric.

| Class        | 2D coverage | Official AP |
|--------------|-------------|-------------|
| Trailer      | 79.5 %      | 0.009       |
| Constr. veh. | 57.5 %      | 0.114       |
| Truck        | 86.8 %      | 0.220       |
| Car          | 90.6 %      | 0.371       |
| Pedestrian   | 77.1 %      | 0.637       |

## 5.2. Coverage by visibility and distance

Table 11 breaks pipeline coverage down along the two axes that dominate it: annotated visibility and range. The measurement here is at the *box level*, whether the pipeline places a box within 4 m of the object, and therefore reflects the full chain from mask to fitted box. Each cell reports the number of ground-truth objects together with their coverage; the total of 121,963 differs marginally from the official count of 121,861 in how objects with zero sensor points are excluded.

The two axes act as independent loss channels. Within every visibility band, coverage falls with range as the LiDAR becomes sparser; within every range band, it falls with occlusion as masks weaken and fewer points remain visible. At the favorable corner—near and fully visible—the pipeline covers 96 % of objects; at the adverse corner it covers 33 %. The training-free pipeline is thus nearly complete where a camera-based system can be expected to operate, and its misses concentrate where the evidence itself vanishes.

## 5.3. Detecting is not scoring

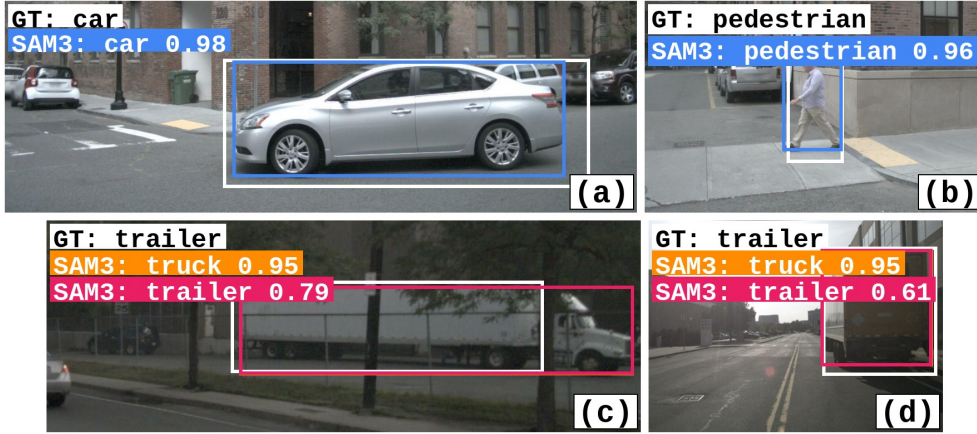
Comparing 2D coverage against official AP exposes a dissociation that class averages hide (Table 12).

Pedestrians convert 77 % coverage into 0.637 AP; trailers convert 79.5 % coverage into 0.009. The trailer failure is therefore not a detection failure: the objects are found, but two mechanisms suppress their score. First, their class is contested: masks on trailers frequently say “truck”, and vice versa, so correct detections are outscored by confidently mislabeled ones. Second, their geometry is unforgiving: on a 12 m articulated body, a center pulled toward the visible face misses the 2–4 m matching thresholds that a pedestrian-sized error survives. The same two mechanisms, in milder form, explain the car–truck gap. Supervised geometry repairs the second mechanism, and trailer AP rises accordingly under borrowed boxes; the class channel persists across every geometry source and caps all mask-classified variants at 0.37–0.40 mAP.

Figure 6 shows both sides of this dissociation. In the top row, a car and a pedestrian each carry a high-scoring mask of the correct class. In the bottom row, two trailers are found by SAM3, but the dominant mask on each says “truck” with score 0.95, while the trailer mask scores lower. The detection exists; its name loses the competition.

## 6. Conclusion

This paper has examined how far open-vocabulary image understanding can be pushed into metric 3D perception without any 3D annotation, through a three-stage comparison in which text-prompted SAM3 segmentation is fixed and only the source of 3D geometry changes. Camera-only lifting through a visual geometry transformer establishes that open-vocabulary detection survives the transfer to 3D, but learned monocular range does not place objects precisely



**Figure 6:** Detection versus naming on nuScenes images [1]. (a) A car and (b) a pedestrian carry a high-scoring mask of the correct class, while (c) and (d) are trailers whose dominant mask says truck.

enough for the strict center-distance metric. Fitting boxes from the raw LiDAR points inside the same masks, with a depth gate and heuristic rules, reaches mAP 0.298 / NDS 0.348 under the official nuScenes protocol at zero labeling cost. Associating supervised box geometry to the same detections at inference time, in the ground plane and, critically, in image space where radial error is invisible, lifts the result to 0.413 / 0.555. Measurement precision is therefore the largest deficit that supervised geometry removes, but not the only limit. Two others survive the substitution: class confusion, where an object is found but carries the wrong name, and confidence calibration, where the scores fail to rank correct detections above incorrect ones. Section 5 shows that the class channel persists across every geometry source, and that an object can be found in the image yet score almost nothing. Reversing the direction, a three-state camera-witness rule built from the same masks lifts a supervised LiDAR-only detector from 0.596 to 0.630 mAP, roughly half the gain of fully supervised camera fusion, with no training.

The accompanying analysis quantifies the 2D detection layer itself. SAM3 covers 84 % of in-range objects with a correctly named mask, and 87.5 % with any mask, and coverage decays along the independent axes of occlusion and range. The classes that fail in the official metric fail not because they are unseen, but because they are misnamed or geometrically unforgiving: trailers are detected in 2D at 79.5 % yet score 0.009 AP. Prompt design emerges as a first-class factor: sub-class prompts measurably improve difficult classes, and prompt wording controls what counts as an object.

One limitation frames every number reported here. The 2D detection component accepts free-text prompts, but the quantitative evaluation is confined to the ten classes of the nuScenes benchmark, so the open-vocabulary claim is measured directly only on those classes. Its wider reach is shown indirectly, by the effect of prompt wording and by detections that are visually correct yet fall outside the evaluated list.

Four directions follow from the analysis. The first is a modular interface between detection and geometry: detection emits a mask, a class name and a score, geometry consumes the mask and returns a box, and this paper shows that three very different back-ends fit behind the same interface. Treating both sides as replaceable modules would turn the comparison into a general evaluation framework for new segmentation models and geometry sources. The second is accumulating LiDAR sweeps. Each box is currently fitted from a single sweep, and sparse returns are the main coverage limit; deciding the existence of an object on several accumulated sweeps, while fitting its geometry on the keyframe sweep alone, is a promising direction. The third is class arbitration from geometry. The dominant confusions (truck–trailer, cone–barrier, truck–construction) are systematic, and the associated box carries strong class evidence: a 12 m box is not a car. Arbitrating the mask-assigned class against the matched box is a training-free path to the class channel that currently caps mask-classified variants. The fourth is moving beyond the fixed class list. Visually correct detections on police vehicles count as false positives here, because the class is excluded from evaluation; prompts that closed-set detectors cannot express, such as *police car*, *ambulance*, *fire truck*, *child*, *stroller* and *road debris*, are exactly where open-vocabulary perception should pay off in driving, and evaluating them requires benchmarks that annotate beyond the standard ten classes.

## CRediT authorship contribution statement

**Ömer Faruk Deniz:** Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Mustafa Taha Koçyiğit:** Supervision, Writing – review and editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The nuScenes dataset is publicly available. Code will be made available on request.

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## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the manuscript text and with typesetting the tables and schematic diagrams from the authors' experimental results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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